

KEVIN SONG

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EDUCATION

VANDERBILT UNIVERSITY

Bachelor of Science, Major in Computer Science, Minor in Business

Nashville, TN

Aug 2023 – May 2027

- **Scholarship:** Full Tuition Merit-Based Ingram Scholarship (awarded to <1% of 1800 students)
- **GPA:** 3.74/4.00
- **Selected Coursework:** Foundations of Machine Learning, Artificial Intelligence, Operating Systems, Database Management Systems, Cybersecurity, Software Engineering, Data Structures, Programming Languages, Computer Architecture

TECHNICAL SKILLS

- **Product & AI Systems:** product design, technical requirements, AI evaluation, retrieval systems, semantic search, vector embeddings, human-centered AI, user-facing workflows
- **Programming & Data:** Python, Java, JavaScript, TypeScript, C, C++, SQL, HTML, CSS; data processing, model validation, inference pipelines
- **Frameworks & Tools:** React, React Native, Node.js, Bun, Supabase, Firebase, Google Cloud Run, PyTorch, YOLO, NumPy, PySide6, Matplotlib, Git, GitHub, Raspberry Pi

PROFESSIONAL EXPERIENCE & PERSONAL PROJECTS

NATIONAL INSTITUTE OF STANDARDS AND TECHNOLOGY (NIST)

Gaithersburg, MD

Software Engineering Intern - Edge AI and Smart Sensors

May 2026 – Aug 2026

- Developing Edge AI smart sensor testbed for ISAC, supporting sensing-assisted wireless reliability and V2X safety research
- Building Python-based AI/ML object detection tools using Raspberry Pi, Blickfeld Qb2 LiDAR, and camera inputs to fuse roadside sensor data, identify pedestrians and vehicles, and support real-time edge perception workflows
- Evaluating RSU/OBU V2X communication latency and pedestrian-safety scenarios to define when roadside perception should trigger early alerts for automated vehicles approaching occluded crossings or high-risk intersections

ELYSIAN AI

New York City, NY

Incoming Software Engineering & Product Management Intern

Aug 2026 – Dec 2026

- Selected for SWE/PM internship focused on ATLAS, an insurance claims-domain knowledge repository powering AI-native claims handling through full-stack development, retrieval systems, and product operations
- Will build database schemas, user-facing interfaces, and retrieval systems for accurate, source-aware AI-agent workflows
- Will bridge Product and Claims by translating claims workflows into structured schemas and AI-agent-ready product surfaces

CANARY.AI (Vanderbilt Senior CS Capstone Project)

Nashville, TN

Full-Stack Mobile AI Project

Jan 2026 – May 2026

- Led 3-member team through product design and full-stack development of React Native/Bun emergency app aggregating local news, location-relevant social posts, and National Weather Service alerts
- Integrated semantic embeddings and Supabase search to filter relevant information and deliver location-specific weather updates
- Reduced time to access critical weather information by 70%; presented at Vanderbilt's CS Senior Symposium

SONGSEARCH (Independent Project)

Clarksburg, MD

AI/ML Music Originality Analysis Platform

Dec 2025 – Mar 2026

- Built cloud-native AI/ML platform analyzing lyrical, rhythmic, motif, and melody to evaluate song originality and copyright risk
- Designed secure upload-to-inference pipeline that extracts audio features, runs originality analysis, and returns structured copyright-risk results for uploaded files in under 30 seconds with minimal user input
- Deployed Firebase database, Google Cloud Run ML inference services, and React/TypeScript frontend, integrating scalable cloud backend workflows with creator-facing music analysis tools

SIGNLECT (Independent Project)

Clarksburg, MD

Computer Vision & Edge AI Accessibility Product

Jul 2025 – Oct 2025

- Built on-device Python computer vision pipeline for real-time American Sign Language translation on Raspberry Pi 5 using camera input, enabling edge AI accessibility support for classroom learning environments
- Trained custom YOLO model using PyTorch and open-source datasets on Google Colab GPUs for accessibility-focused use cases
- Deployed and optimized model on constrained hardware, improving inference speed by 100% for low-cost edge AI translation

ADDITIONAL INFORMATION

- **Languages:** English (Native), Mandarin Chinese (Native), Spanish (Conversational)
- **Interests:** Engineering competitions, songwriting and music production, tennis, piano